

Project Plan

Project Title: *E-Commerce Order Journey Analysis Using the Olist Dataset*

Team Name: *Four People and Some Data*

Date: *May 27, 2025*

I. Introduction

This capstone project leverages the publicly available Olist E-commerce dataset from a Brazilian online marketplace. The dataset spans the full lifecycle of customer orders—from placement to delivery and feedback—and merges data from multiple tables including customers, orders, sellers, products, reviews, and payments.

With over 119,000 entries and 40 features, the dataset enables deep analysis into customer satisfaction, delivery performance, seller behavior, and regional dynamics. Our focus is to explore delivery bottlenecks, seller behavior, and customer segmentation to derive insights that enhance platform reliability and user experience.

II. Business Description

Olist is a Brazilian e-commerce company that provides a platform for small and medium-sized businesses (SMBs) to sell their products across major online marketplaces in Brazil. Founded in 2015 and headquartered in Curitiba, Brazil, Olist helps merchants by offering tools for product listing, inventory management, logistics, and customer service.

Although Olist is currently a private company, it has received significant venture capital funding and is considered one of Brazil's leading e-commerce enablers.

As of the latest public reports:

- Annual Revenue (estimated): Over USD 100 million
- Total Funding Raised: Over USD 100 million from investors including Goldman Sachs, SoftBank, and Valor Capital Group

Key Executives:

- Founder & CEO: Tião Rezende (Tião Rezende de Araújo)
- Chief Operating Officer: (Position may vary depending on the latest org chart)

Main Competitors:

- Loja Integrada
- Tray
- B2W Digital

- VTEX

Olist operates by aggregating thousands of merchants and listing their products on large marketplaces like Amazon Brazil, Mercado Livre, Americanas, and Magazine Luiza, streamlining the e-commerce process for businesses that might otherwise lack the tools or visibility to sell online effectively.

III. Project Objectives

We aim to:

- Identify predictors of late deliveries and flag high-risk orders for proactive handling.
 - Detect seller-level delays and understand how upstream issues contribute to customer-facing problems.
 - Analyze customer segments for return and cancellation behaviors to anticipate and reduce churn.
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IV. Research Questions

1. What key factors lead to late deliveries, and how can we proactively identify high-risk orders to improve customer satisfaction?

Hypothesis: There are categorical variables that consistently predict late deliveries

2. Which sellers consistently delay order handoffs to carriers, and what underlying patterns or behaviors contribute to this? Can early seller delays help us anticipate customer-facing delivery issues?

Hypothesis: There are categorical and/or distance-related variables within order data that will predict lateness among sellers that consistently deliver late to carriers.

3. How do customer segments—such as first-time vs. repeat buyers or by region—differ in cancellation and return behavior, and can we predict these actions early based on browsing or shopping patterns to reduce churn?

Hypothesis: There are categorical and/or behavioral variables (like buyer type, region, browsing time, cart activity) that consistently predict cancellation and return behavior across customer segments.

V. Data Description

We are using the full Olist dataset, composed of:

- **Customer-level Data:** Location, unique ID, segmentation

- **Order Information:** Status, purchase time, approval time, delivery estimates, actual delivery
- **Sellers:** Seller location and ID
- **Products:** Category (with English translation), price, dimensions
- **Payments:** Method, number of installments, value
- **Reviews:** Score, comments, review timestamp

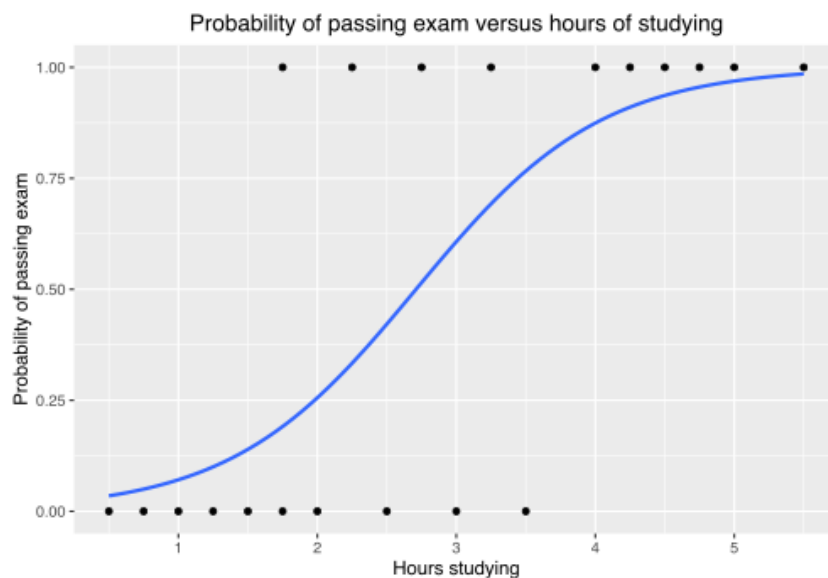
The dataset is already joined as `olist_combined_dataset.csv`, and includes translated product categories for improved analysis.

VI. Methodology

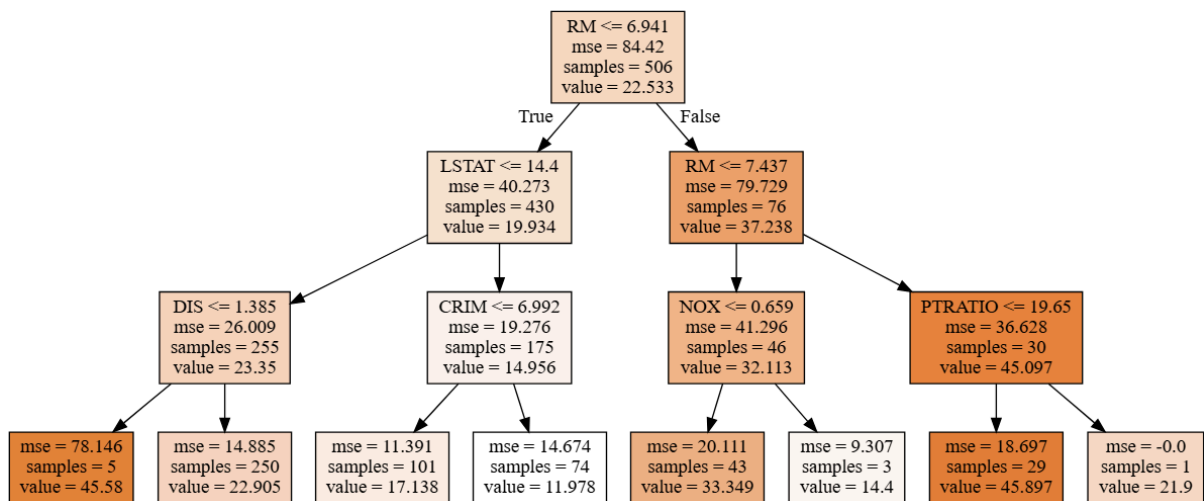
Each research question will be tackled using a tailored approach:

Q1 – Late Delivery Prediction

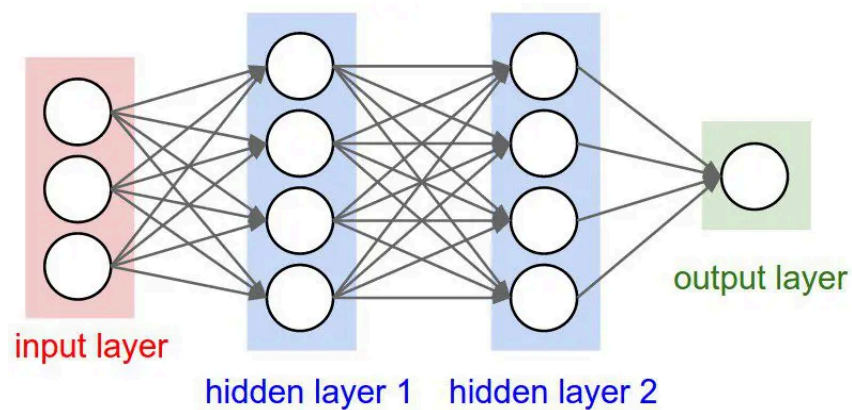
- Label orders as late if Delivered Date > Estimated Delivery Date
- Feature engineering: shipping gap, product weight/volume, distance between seller and customer, etc.
- Train classifiers: Logistic Regression, Decision Tree, Random Forest, XGBoost, Artificial Neural Network (ANN)
- Evaluate with ROC AUC, Precision, Recall, F1
- Output probability scores to flag high-risk orders



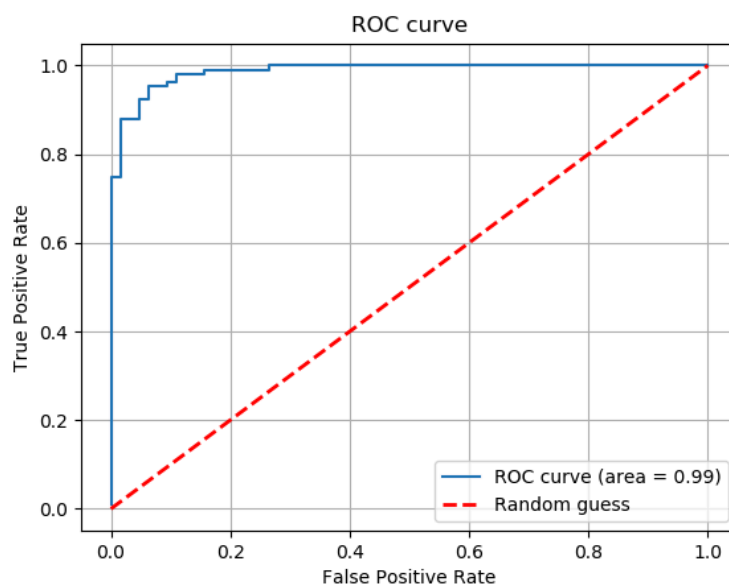
Example of Logistic Regression Graph. Similar Method will be used for working on RQ1



Example of Decision Tree Graph. Similar Method will be used for working on RQ1



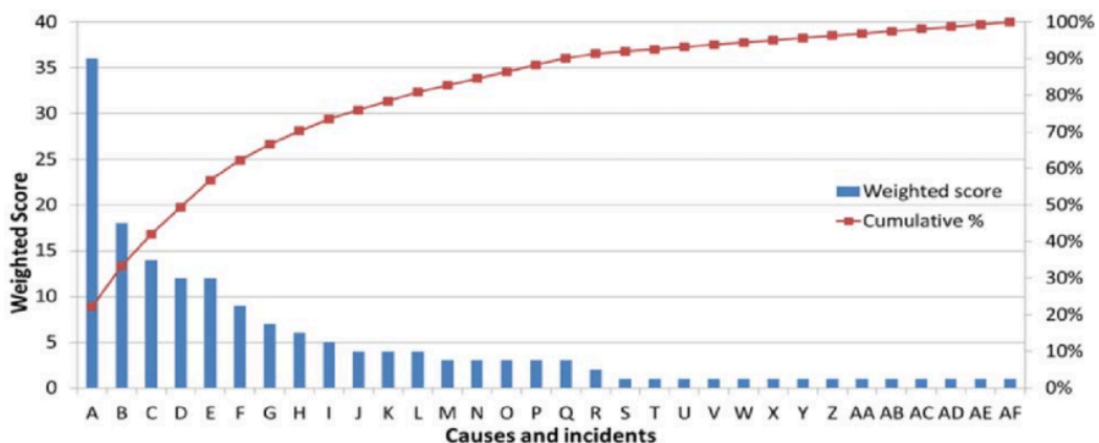
Example of Artificial Neural Network that will be used for working on RQ1



Example of ROC Curve with AUC. Similar method will be used to evaluate different models created to work on RQ1

Q2 – Seller Delay and Impact

- Compare shipping_limit_date to order_delivered_carrier_date to find late deliveries to the carrier
- Correlation analysis between late to carrier and late to customer orders
 - Regression (possibly with data transformation) to identify if there's a predictable relationship between late to carrier and late to customer conditions
- From late deliveries, use Pareto Principle to determine suppliers to focus on (top 20% of suppliers that cause 80% of the overall late deliveries)



Example of Pareto chart showing how to use counts/weights and cumulative % (typically 80% level) to choose focus for further investigation (Bergli, et al., 2017)

- Multilinear regression with feature reduction (involving seller, order, and product data) to determine what factors, if any, allow useful prediction of late to carrier events within the suppliers of interest

OLS Regression Results						
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Dep. Variable:	R	R-squared:	0.943			
Model:	OLS	Adj. R-squared:	0.943			
Method:	Least Squares	F-statistic:	2965.			
Date:	Fri, 06 Dec 2024	Prob (F-statistic):	0.00			
Time:	01:47:08	Log-Likelihood:	-6751.1			
No. Observations:	1432	AIC:	1.352e+04			
DF Residuals:	1423	BIC:	1.357e+04			
DF Model:	8					
Covariance Type:	nonrobust					
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	coef	std err	t	P> t	[0.025	0.975]
Intercept	-27.6145	6.113	-4.517	0.000	-39.607	-15.622
H	0.2264	0.008	27.028	0.000	0.210	0.243
doub	0.3455	0.030	11.547	0.000	0.287	0.404
trip	0.7798	0.083	9.375	0.000	0.617	0.943
HR	1.2227	0.025	49.550	0.000	1.174	1.271
BB	0.2723	0.011	23.778	0.000	0.250	0.295
SO	-0.0849	0.005	-18.645	0.000	-0.094	-0.076
HBP	0.4351	0.055	7.968	0.000	0.328	0.542
SF	1.0264	0.099	10.393	0.000	0.833	1.220
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Omnibus:	12.025	Durbin-Watson:	1.484			
Prob(Omnibus):	0.002	Jarque-Bera (JB):	13.763			
Skew:	0.152	Prob(JB):	0.00103			
Kurtosis:	3.372	Cond. No.	1.58e+04			
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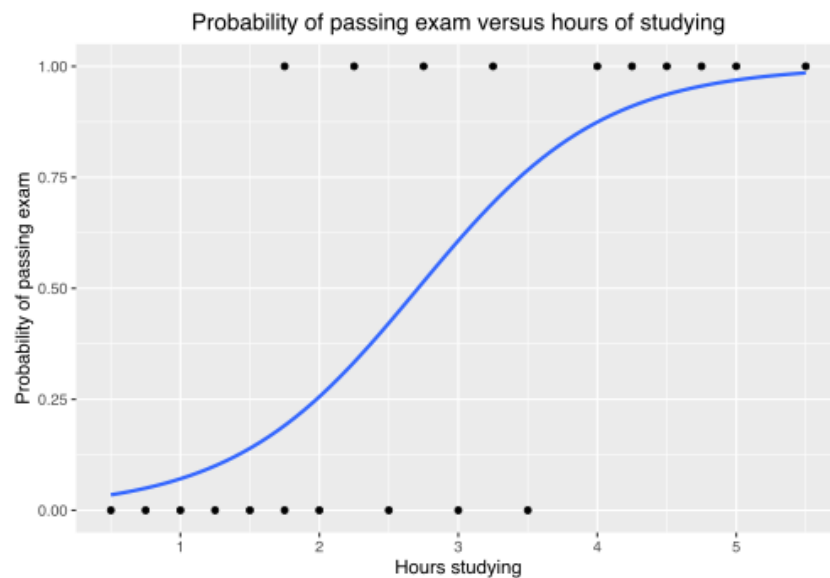
Example multilinear regression output showing variable coefficients and evaluation of

statistical significance for each coefficient. (Opitz, 2024)

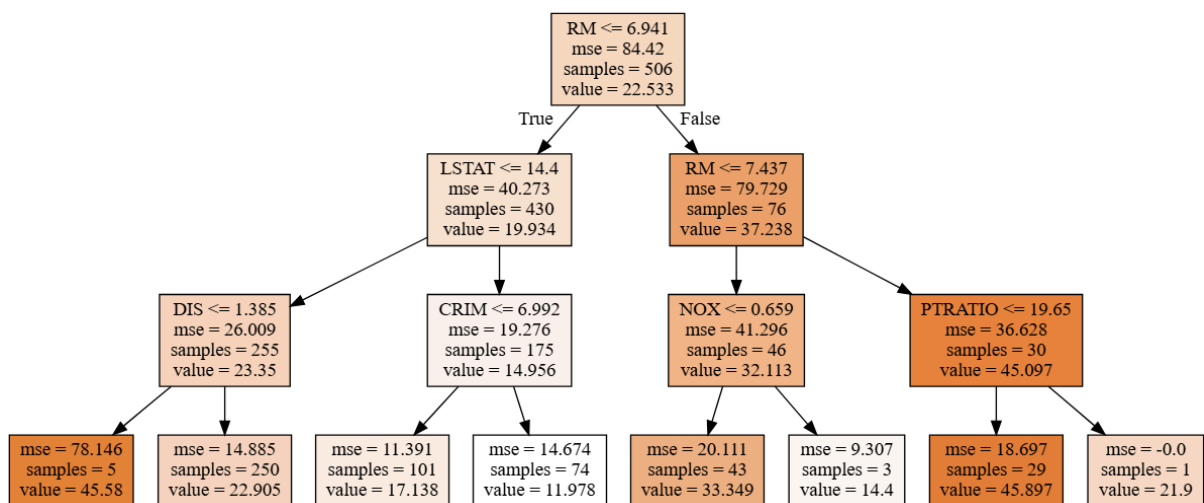
- Highlight largest risk areas by supplier for handoff to supplier teams to determine root causes

Q3 – Customer Segmentation and Churn Prediction

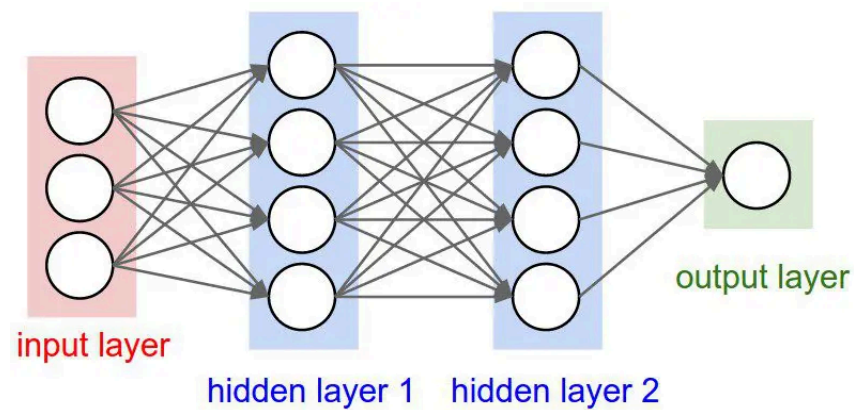
- Two parts methodology:
 1. group comparison (statistical analysis)
 2. behavioral prediction (machine learning)
- For group comparison, separate customers by different groups, use ANOVA test and bar plots for data visualization.
- For behavioral prediction, use binary variable, try logistic regression, XGBoost, and/or create a neural network, then test the accuracy, precision, recall, and ROC-AUC



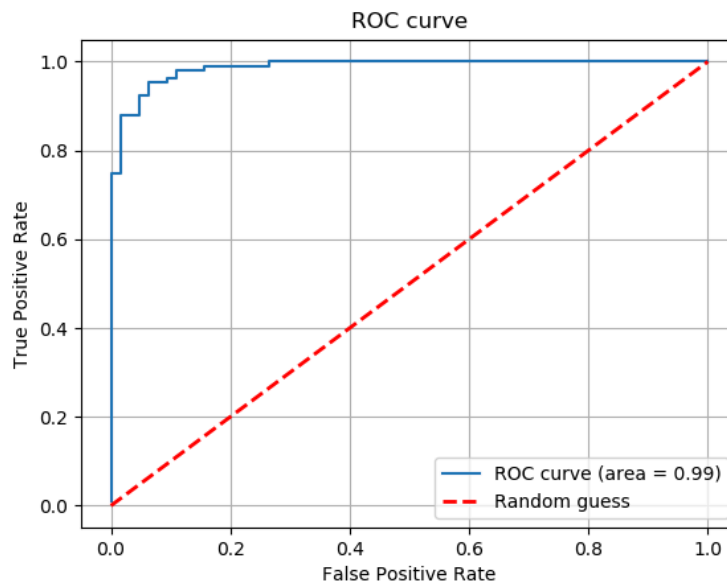
Example of Logistic Regression Graph. Similar Method will be used for working on RQ3



Example of Decision Tree Graph. Similar Method will be used for working on RQ3



Example of Artificial Neural Network that will be used for working on RQ3



Example of ROC Curve with AUC. Similar method will be used to evaluate different models created to work on RQ3

VII. Project Timeline

Week	Activities / Deliverables
Week 1	Project Planning, Literature Review

Week 2	Dataset Familiarization, EDA, Dataset Cleaning
Week 3	Dataset Analysis, Creating Models and Visualizations, Ethical Recommendations
Week 4	Rough Draft
Week 5	Final Draft
Week 6	Video Presentations, Summative Feedback Papers

VIII. Roles and Responsibilities

Team Member	Roles
Meshach Samuel	Perform Data Cleaning, Work on Models for RQ1, Ethical Recommendations
Yanzhe Qu	Work on Models for RQ3, related analysis with visualizations
Matt Opitz	Literature Review, RQ2 Preliminary Analysis and Follow-on Analysis with Visualizations, Presentation Lead
Suyong Choi	EDA

All members will participate in weekly check-ins, reviews, and presentation prep.

IX. Tools and Resources

- **Languages:** Python (pandas, NumPy, scikit-learn, LightGBM)
 - **Visualization:** matplotlib, seaborn, Plotly
 - **Collaboration:** GitHub, Google Drive, Text Messages
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X. Anticipated Challenges

- **Missing timestamps** in records requiring imputation or row filtering
 - **Imbalanced labels** for rare events like late deliveries or cancellations
 - **Multi-table joins** with potential data consistency issues
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XI. Final Deliverables

- Predictive models for late deliveries and churn
 - Seller performance analysis with supporting visualizations
 - Segment-level behavior analysis and risk scoring
 - Final written report (~10 pages)
 - Slide deck for oral presentation
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XII. References

Bergli, Simen & Falk, Kristin. (2017). "Cause and Impact Analysis of Cost and Schedule Overruns in Subsea Oil and Gas Projects – A Supplier's Perspective." *INCOSE International Symposium*. **27**, 309-321. <https://doi.org/10.1002/j.2334-5837.2017.00362.x>.

M. Opitz. "DAT 301 - 76300 - Project 2". Arizona State University DAT 301, 2024 Fall B Session, 12/4/2024.